

Experience

- 2025–Now **Senior AI Research Scientist**,
Noah's Ark Lab, Huawei Technologies Canada, Toronto, Canada
- * Developing audio foundation models (Large Audio-Language Models) for end-to-end speech-to-speech conversation systems
 - * Designing agentic AI pipelines for automated audio editing using LALMs and tool-use frameworks
- 2022–2025 **AI Research Scientist**,
LG Electronics AI Lab, Toronto, Canada
- * Leveraged LLMs and generative models to synthesize training data for low-resource spoken language understanding tasks, reducing annotation cost
 - * Built multimodal perception systems for smart home environments, combining audio, vision, and language signals
 - * Developed deep learning models for audio event detection and speech understanding deployed in LG smart devices
 - * Estimated room impulse responses and acoustic properties from visual room images using cross-modal learning
- 2021–2022 **Machine Learning Engineer**,
IMRSV Data Labs, Ottawa, Canada
- * Researched and built an end-to-end speech-to-speech translation system using sequence-to-sequence architectures
 - * Optimized and deployed automated speech recognition models on mobile devices under strict memory and latency constraints
- 2016–2018 **Senior Software Engineer**,
AEC, Tehran, Iran
- * Led a team of 4 engineers to design and deliver RESTful web APIs using Django and Flask
 - * Built an automation platform for cellphone coverage tracking, reducing manual reporting effort significantly

Education

- 2018–2022 **Ph.D. Artificial Intelligence**,
Queen's University
- Worked on audio-based deep learning solutions for smart meeting rooms with a focus on deep speaker representation learning. The speaker representations can be used for speaker identification and emotion recognition during the meetings which helps the meeting organizers to have: a) a better automated minute extraction system b) a better sense of the emotional/cognitive state of the attendees so that they can organize the meetings more efficiently
- 2014–2016 **M.Sc. Computer Science**,
University of Tehran
- Used a combination of language models and finite-state transducers to develop a correction model for white space misplacement. Using a finite-state transducer with weights assigned from language models, the proposed solution was able to accurately correct the misplaced white spaces in cases where the separated sub-words could be considered meaningful words
- 2008–2013 **B.Sc. Computer Engineering**,
Zanjan University
- Implemented an inefficient data-gathering and reporting web service using python for tenure candidate assessment. By optimizing the dataset structure and queries, I was able to drastically enhance the performance of data-gathering and reporting system. I further improved the system by adding an automated publication registration using crawler agents.

Tools and Skills

Programming Languages: Python, Bash

Deep Learning Libraries: PyTorch, TensorFlow, HuggingFace Transformers

Speech & Audio: Kaldi, ESPnet, Whisper, WavLM, HuBERT, Librosa

Agentic AI: Pydantic, LangChain, SmolAgents, LangGraph

Methods & Tools: Agile, Git, Docker, CUDA

Platforms: Google Cloud Platform, AWS, Sun Grid Engine

Selected Publications

489 citations · h-index: 7 · i10-index: 7

- 2019 **A. Hajavi**, A. Etemad. *A deep neural network for short-segment speaker recognition*. Interspeech.
- 2020 Kourkounakis, **A. Hajavi**, A. Etemad. *Detecting multiple speech disfluencies using a deep residual network with BiLSTM*. ICASSP.
- 2021 Kourkounakis, **A. Hajavi**, A. Etemad. *FluentNet: End-to-end detection of stuttered speech disfluencies*. IEEE/ACM TASLP.
- 2021 **A. Hajavi**, A. Etemad. *Siamese capsule network for end-to-end speaker recognition in the wild*. ICASSP.
- 2023 **A. Hajavi**, A. Etemad. *Fine-grained early frequency attention for deep speaker representation learning*. IEEE Trans. AI.
- 2023 **A. Hajavi**, A. Etemad. *A study on bias and fairness in deep speaker recognition*. ICASSP.
- 2024 **A. Hajavi** et al. *Joint vs. unimodal representation learning for audio-visual emotion recognition*. IJCNN.
- 2025 S. Jia, **A. Hajavi** et al. *ID-TTA: Classifier-free test time adaptation for metric learning*. ICIP.